

IRON STEEL & METAL PRODUCTS

CHUNK 1

Product:

Galvanized Steel Sheets and Strips

Description:

Galvanized steel sheets and strips are steel products coated with zinc to improve resistance to atmospheric corrosion. They are widely used in roofing, construction, fabrication, ducts, panels and other applications where corrosion protection is required.

Applicable IS:

IS 277:2003

Standard title:

Galvanized Steel Sheets (Plain and Corrugated) — Specification

Scope / applicability:

The standard covers galvanized steel sheets in plain and corrugated forms, subject to the dimensional, coating, grade and other requirements specified in the standard.

BIS certification:

Mandatory for products falling within the applicable Quality Control Order.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark.

Applicable QCO:

Steel and Steel Products (Quality Control) Order, 2024, as amended, subject to the product's exact scope and enforcement provisions.

Current status — August 2026:

Listed by BIS under Steel and Iron Products subject to compulsory certification.

Product Manual:

Applicable BIS Product Manual for IS 277.

SIT:

Product-specific Scheme of Inspection and Testing covering manufacturing controls, sampling, testing frequency, acceptance criteria and marking.

Required testing:

- Chemical composition
- Zinc coating/coating mass
- Tensile properties
- Yield strength where applicable
- Elongation
- Bend test
- Dimensions
- Thickness
- Surface condition
- Coating uniformity/adhesion as specified
- Mass-related requirements
- Other tests specified by the applicable Product Manual

Manufacturing / quality requirements:

The manufacturer needs appropriate production controls, testing facilities and documented quality-control procedures for the grades and sizes covered by the licence.

Documents / certification requirements:

- BIS licence application
- Relevant Indian Standard
- Manufacturing-process details
- Quality-control plan
- In-house testing capability
- Calibration records
- Test records
- Applicable Product Manual/SIT
- Factory assessment requirements
- Third-party/manufacturer testing as applicable

Marking requirements:

BIS Standard Mark / ISI Mark and applicable identification, grade, dimensions and manufacturer/licence information as prescribed.

Important exclusions:

The exact applicability depends on the product form, dimensions, coating and intended use specified in the QCO/standard.

2026 special note:

The BIS compulsory-certification list currently identifies IS 277:2003 under Steel and Iron Products. The applicable QCO and amendments must be checked for the exact product before determining enforcement obligations.

Official BIS source:

BIS — Products Under Compulsory Certification

CHUNK 2

Product:

Mild Steel Wire for General Engineering Purposes

Description:

Mild steel wire is a low-carbon steel wire used in general engineering, fabrication, tying and other industrial applications. Its mechanical properties and dimensional accuracy are important for reliable performance.

Applicable IS:

IS 280:2006

Standard title:

Mild Steel Wire for General Engineering Purposes — Specification

Scope / applicability:

Covers mild steel wire intended for general engineering purposes within the grades, sizes and conditions specified by the standard.

BIS certification:

Mandatory where covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark.

Applicable QCO:

Steel and Steel Products (Quality Control) Order framework, subject to current amendments and exact scope.

Current status — August 2026:

Listed by BIS among steel and iron products under compulsory certification.

Product Manual:

Applicable Product Manual for IS 280.

SIT:

Applicable Scheme of Inspection and Testing covering sampling, mechanical testing, chemical requirements and routine quality control.

Required testing:

- Chemical composition
- Tensile strength
- Elongation
- Diameter
- Dimensional tolerances
- Surface condition
- Bend/other mechanical tests where specified
- Freedom from harmful defects
- Other tests prescribed by the standard/SIT

Manufacturing / quality requirements:

Controlled steelmaking, wire drawing and finishing processes appropriate to the grades and sizes covered by the licence.

Documents / certification requirements:

- BIS application
- Indian Standard
- Manufacturing-process documentation
- QC plan
- Test facilities
- Calibration records
- Test records
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark with required identification and licence details.

Important exclusions:

Special-purpose wires may be governed by separate Indian Standards.

2026 special note:

Do not map every product described commercially as "steel wire" to IS 280. The intended application and technical specification must first be identified.

Official BIS source:

BIS — Scheme-I Steel and Iron Products

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/scheme-i-mark-scheme/>

CHUNK 3

Product:

Expanded Metal Steel Sheets

Description:

Expanded metal sheets are manufactured by simultaneously slitting and stretching metal sheet to form a regular mesh pattern. They are used for guards, partitions, walkways, screens, ventilation and structural applications.

Applicable IS:

IS 412:1975

Standard title:

Expanded Metal Steel Sheets for General Purposes — Specification

Scope / applicability:

Steel expanded metal sheets intended for general purposes within the dimensional and material requirements specified by the standard.

BIS certification:

Mandatory where covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark.

Applicable QCO:

Steel and Steel Products (Quality Control) Order framework.

Current status — August 2026:

Listed in the BIS compulsory-certification material.

Product Manual:

Applicable Product Manual for IS 412.

SIT:

Product-specific sampling and testing requirements.

Required testing:

- Chemical composition
- Sheet thickness
- Mesh dimensions
- Strand dimensions
- Length/width
- Mechanical properties where specified
- Surface condition
- Dimensional tolerances
- Other standard-specific requirements

Manufacturing / quality requirements:

The manufacturing process must provide consistent mesh geometry, dimensions and material properties.

Documents / certification requirements:

- BIS application
- Relevant IS
- Manufacturing details
- QC procedures
- Testing facilities
- Calibration records
- Test records
- Product Manual/SIT

- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark and prescribed product identification.

Important exclusions:

Expanded products made from materials other than the steel grades covered by the standard may fall outside this record.

2026 special note:

The AI should distinguish expanded metal from perforated sheet, wire mesh and welded mesh.

Official BIS source:

BIS — Products Under Compulsory Certification

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/>

CHUNK 4

Product:

Mild Steel and Medium Tensile Steel Bars for Concrete Reinforcement

Description:

These are steel reinforcement bars used inside concrete structures to provide tensile strength and improve structural performance. They are critical construction products because their mechanical and chemical properties directly affect structural safety.

Applicable IS:

IS 432 (Part 1):1982

Standard title:

Mild Steel and Medium Tensile Steel Bars and Hard-Drawn Steel Wire for Concrete Reinforcement — Part 1: Mild Steel and Medium Tensile Steel Bars

Scope / applicability:

Bars intended for reinforcement of concrete structures within the grades, dimensions and mechanical requirements specified by the standard.

BIS certification:

Mandatory where covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark.

Applicable QCO:

Steel and Steel Products (Quality Control) Order framework.

Current status — August 2026:

Listed by BIS under compulsory steel and iron certification.

Product Manual:

Applicable BIS Product Manual for IS 432 Part 1.

SIT:

Includes controls for chemical composition, mechanical properties, dimensions and routine production testing.

Required testing:

- Chemical composition
- Yield stress
- Tensile strength
- Elongation
- Bend test
- Re-bend where applicable
- Mass per metre
- Diameter/dimensions
- Surface condition
- Freedom from harmful defects

Manufacturing / quality requirements:

Steelmaking, rolling and finishing processes must be controlled to consistently achieve the specified mechanical and chemical requirements.

Documents / certification requirements:

- BIS licence application
- Manufacturing-process details
- Grade details
- QC plan
- Test facilities
- Calibration
- Routine test records
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark and required grade/manufacturer/licence identification.

Important exclusions:

Higher-strength reinforcement products governed by newer standards should not automatically be classified under IS 432.

2026 special note:

For structural-safety queries, the AI should ask for the exact reinforcement grade and standard rather than responding only from the phrase "TMT bar."

Official BIS source:

BIS — Scheme-I Steel and Iron Products

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/scheme-i-mark-scheme/>

CHUNK 5

Product:

Hard-Drawn Steel Wire for Concrete Reinforcement

Description:

Hard-drawn steel wire is manufactured by drawing steel through dies to achieve the required diameter and mechanical properties. It can be used as reinforcement material in concrete applications.

Applicable IS:

IS 432 (Part 2):1982

Standard title:

Mild Steel and Medium Tensile Steel Bars and Hard-Drawn Steel Wire for Concrete Reinforcement — Part 2: Hard-Drawn Steel Wire

Scope / applicability:

Hard-drawn steel wire intended for concrete reinforcement within the sizes, grades and requirements specified by the standard.

BIS certification:

Mandatory where covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark.

Applicable QCO:

Steel and Steel Products (Quality Control) Order framework.

Current status — August 2026:

Listed by BIS under compulsory certification.

Product Manual:

Applicable Product Manual for IS 432 Part 2.

SIT:

Product-specific mechanical, dimensional and chemical testing requirements.

Required testing:

- Chemical composition
- Tensile strength
- Yield-related properties where specified
- Elongation
- Diameter

- Dimensional tolerances
- Surface condition
- Bend-related requirements where specified
- Mass-related requirements
- Other standard-specific tests

Manufacturing / quality requirements:

Controlled wire drawing, heat treatment where applicable, finishing and inspection processes.

Documents / certification requirements:

- BIS application
- Manufacturing details
- QC plan
- Test equipment
- Calibration
- Test records
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark with required identification.

Important exclusions:

Prestressing wires and other specialized steel wires may fall under different standards.

2026 special note:

Do not treat every concrete-reinforcement wire as IS 432 Part 2. Prestressed-concrete wire is a separate technical category.

Official BIS source:

BIS — Products Under Compulsory Certification

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/>

CHUNK 6

Product:

Cold Reduced Carbon Steel Sheets and Strips — Cold Forming and Drawing
Purpose

Description:

Cold-reduced carbon steel sheets and strips are steel products manufactured by reducing hot-rolled material at room temperature. They are widely used in automotive, appliance, fabrication and forming applications.

Applicable IS:

IS 513 (Part 1):2016

Standard title:

Cold Reduced Carbon Steel Sheets and Strips — Part 1: Cold Forming and Drawing
Purpose

Scope / applicability:

Cold-reduced carbon steel sheets and strips intended for cold forming and drawing applications within the specified grades and dimensions.

BIS certification:

Mandatory where covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark.

Applicable QCO:

Steel and Steel Products (Quality Control) Order framework.

Current status — August 2026:

Listed under BIS compulsory steel certification.

Product Manual:

Applicable Product Manual for IS 513 Part 1.

SIT:

Product-specific chemical, mechanical, dimensional and surface-quality testing.

Required testing:

- Chemical composition
- Yield strength
- Tensile strength
- Elongation
- Plastic strain ratio where specified
- Surface quality
- Thickness
- Width
- Flatness
- Dimensional tolerances
- Bend/formability-related requirements
- Other tests specified by the standard

Manufacturing / quality requirements:

Controlled rolling, annealing, skin passing and finishing processes as applicable to the grade.

Documents / certification requirements:

- BIS application
- Manufacturing-process details
- Grade/product range
- QC plan
- Laboratory facilities
- Calibration records
- Test records
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark and prescribed product identification.

Important exclusions:

High-tensile and multi-phase cold-reduced steels are separately classified under IS 513 Part 2.

2026 special note:

The AI should distinguish Part 1 from Part 2 rather than treating all cold-reduced carbon steel under one standard.

Official BIS source:

BIS — Scheme-I Steel and Iron Products

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/scheme-i-mark-scheme/>

CHUNK 7

Product:

Cold Reduced High Tensile and Multi-Phase Steel Sheets and Strips

Description:

These are advanced cold-reduced steel products designed to provide higher strength and specialized forming characteristics. They are important in automotive and lightweight structural applications.

Applicable IS:

IS 513 (Part 2):2016

Standard title:

Cold Reduced Carbon Steel Sheets and Strips — Part 2: High Tensile and Multi-Phase Steel

Scope / applicability:

High-tensile and multi-phase cold-reduced steel sheets and strips within the grades and dimensions specified by the standard.

BIS certification:

Mandatory where covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark.

Applicable QCO:

Steel and Steel Products (Quality Control) Order framework.

Current status — August 2026:

Listed under compulsory BIS steel certification.

Product Manual:

Applicable Product Manual for IS 513 Part 2.

SIT:

Product-specific testing for chemical, mechanical and dimensional requirements.

Required testing:

- Chemical composition
- Yield strength
- Tensile strength
- Elongation
- Formability-related properties
- Thickness
- Width
- Flatness
- Surface quality
- Dimensional tolerances
- Other grade-specific tests

Manufacturing / quality requirements:

Controlled steelmaking and rolling/processing conditions suitable for the specified high-strength or multi-phase grade.

Documents / certification requirements:

- BIS application
- Manufacturing process
- Grade details
- Testing capability
- Calibration
- Test records
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark with prescribed product identification.

Important exclusions:

Ordinary cold-forming grades belong to Part 1.

2026 special note:

The AI should recognize terms such as "AHSS" only as a search clue and verify the actual BIS grade/standard before assigning a compliance requirement.

Official BIS source:

BIS — Scheme-I Steel and Iron Products

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/scheme-i-mark-scheme/>

CHUNK 8

Product:

Cold Rolled Non-Oriented Electrical Steel Sheets and Strips — Fully Processed Type

Description:

Cold rolled non-oriented electrical steel, commonly called CRNO steel, is used primarily in electrical machines such as motors, generators and transformers where magnetic properties and low core losses are important.

Applicable IS:

IS 648:2006

Standard title:

Cold Rolled Non-Oriented Electrical Steel Sheets and Strips — Fully Processed Type

Scope / applicability:

Fully processed non-oriented electrical steel sheets and strips intended for electrical and magnetic applications within the grades specified by the standard.

BIS certification:

Mandatory where covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark.

Applicable QCO:

Steel and Steel Products (Quality Control) Order framework.

Current status — August 2026:

Listed by BIS under compulsory steel certification.

Product Manual:

Applicable Product Manual for IS 648.

SIT:

Product-specific electrical, magnetic, chemical, mechanical and dimensional testing.

Required testing:

- Magnetic induction
- Core loss
- Electrical resistivity
- Thickness
- Flatness
- Chemical composition
- Mechanical properties where specified
- Surface insulation characteristics where specified
- Dimensional tolerances
- Other magnetic-property tests

Manufacturing / quality requirements:

Controlled steel composition, rolling, annealing and surface treatment are important for achieving specified magnetic performance.

Documents / certification requirements:

- BIS application
- Manufacturing process
- Grade/product range
- Laboratory facilities
- Calibration records

- Magnetic testing capability
- Test records
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark and required grade/product identification.

Important exclusions:

Grain-oriented electrical steel is a different technical category.

2026 special note:

The assistant should distinguish ****CRNO**** from ****CRGO**** steel when answering industry queries.

Official BIS source:

BIS — Products Under Compulsory Certification

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/>

CHUNK 9

Product:

Chrome Molybdenum Steel Bars and Rods for Aircraft Purposes

Description:

Chrome molybdenum alloy steel bars and rods provide enhanced strength and performance for specialized engineering applications. The aircraft-purpose category has stringent material and mechanical requirements.

Applicable IS:

IS 963:1958

Standard title:

Chrome Molybdenum Steel Bars and Rods for Aircraft Purposes — Specification

Scope / applicability:

Chrome molybdenum steel bars and rods intended for aircraft applications within the specified grades and dimensions.

BIS certification:

Mandatory where covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark.

Applicable QCO:

Steel and Steel Products (Quality Control) Order framework.

Current status — August 2026:

Listed in BIS compulsory steel and iron certification material.

Product Manual:

Applicable BIS Product Manual for IS 963.

SIT:

Product-specific chemical and mechanical testing.

Required testing:

- Chemical composition
- Tensile strength
- Yield strength
- Elongation
- Hardness where specified
- Heat-treatment condition
- Dimensions
- Surface quality
- Other grade-specific requirements

Manufacturing / quality requirements:

Controlled alloy steelmaking, heat treatment and finishing processes appropriate to aircraft-grade material.

Documents / certification requirements:

- BIS application
- Material grade details
- Manufacturing process
- Heat-treatment records
- Laboratory capability
- Calibration
- Test records
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark with required grade, batch and manufacturer identification.

Important exclusions:

Aircraft components and finished aerospace parts are not automatically covered by this material standard.

2026 special note:

The AI should distinguish the ****material standard**** from certification of the final aircraft component.

Official BIS source:

BIS — Scheme-I Steel and Iron Products

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/scheme-i-mark-scheme/>

CHUNK 10

Product:

Hot Rolled Steel Strip for Baling

Description:

Hot rolled steel strip for baling is a steel strip product used for bundling, securing and packaging heavy materials. Its strength, dimensions and surface condition are important for safe handling and load retention.

Applicable IS:

IS 1029:1970

Standard title:

Specification for Hot Rolled Steel Strip (Baling)

Scope / applicability:

Hot rolled steel strip intended for baling applications within the grades, dimensions and mechanical requirements specified by the standard.

BIS certification:

Mandatory where covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark.

Applicable QCO:

Steel and Steel Products (Quality Control) Order framework.

Current status — August 2026:

Listed in BIS steel and iron compulsory-certification material.

Product Manual:

Applicable BIS Product Manual for IS 1029.

SIT:

Product-specific chemical, mechanical and dimensional testing.

Required testing:

- Chemical composition
- Tensile strength
- Elongation
- Thickness
- Width
- Dimensional tolerances

- Surface condition
- Bend-related properties where specified
- Other standard-specific requirements

Manufacturing / quality requirements:

Controlled hot rolling and finishing processes are required to maintain consistent strip properties.

Documents / certification requirements:

- BIS application
- Manufacturing process details
- QC plan
- Testing capability
- Calibration
- Test records
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark with prescribed manufacturer, grade and product identification.

Important exclusions:

Other hot rolled steel strips intended for different applications may be governed by different standards.

2026 special note:

The AI should identify the intended use when a user simply says "steel strip," because multiple BIS standards may apply.

Official BIS source:

BIS — Scheme-I Steel and Iron Products

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/scheme-i-mark-scheme/>

CHUNK 11

Product:

Hot Rolled Low Carbon Steel Sheets and Strips for Cold Forming

Description:

Hot rolled low-carbon steel sheets and strips are produced by rolling steel at elevated temperatures and are used in forming, fabrication, automotive and general engineering applications. Their chemical composition, mechanical properties and dimensional accuracy are important for subsequent processing.

Applicable IS:

IS 1079:2017

Standard title:

Hot Rolled Carbon Steel Sheet and Strip — Specification

Scope / applicability:

Covers hot rolled carbon steel sheets and strips intended for general engineering and forming applications within the grades, dimensions and delivery conditions specified by the standard.

BIS certification:

Mandatory where the product falls within the applicable Steel and Steel Products Quality Control Order.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark.

Applicable QCO:

Steel and Steel Products (Quality Control) Order, 2024, as amended.

Current status — August 2026:

Subject to compulsory BIS certification where covered by the current QCO scope.

Product Manual:

Applicable BIS Product Manual for IS 1079.

SIT:

Product-specific Scheme of Inspection and Testing covering chemical composition, mechanical properties, dimensions, surface quality and routine production controls.

Required testing:

- Chemical composition
- Yield strength where specified
- Tensile strength
- Elongation
- Bend test
- Thickness
- Width
- Flatness
- Dimensional tolerances
- Surface condition
- Freedom from harmful defects

- Other grade-specific requirements

Manufacturing / quality requirements:

Controlled steelmaking, hot rolling, cooling and finishing processes are required to maintain the specified grade and mechanical properties.

Documents / certification requirements:

- BIS application
- Relevant Indian Standard
- Manufacturing-process details
- Grade/product range
- Quality-control plan
- Testing facilities
- Calibration records
- Routine test records
- Product Manual/SIT
- Factory assessment documents

Marking requirements:

BIS Standard Mark / ISI Mark with prescribed grade, size, manufacturer and licence identification.

Important exclusions:

Cold rolled steel products, galvanized products and specialized high-strength steels may be governed by separate standards.

2026 special note:

The AI should not map every product described as "hot rolled steel sheet" to one standard without checking grade, thickness, intended use and applicable QCO scope.

Official BIS source:

BIS — Products Under Compulsory Certification

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/>

CHUNK 12

Product:

Hot Rolled Medium and High Tensile Structural Steel

Description:

Structural steel is used in buildings, bridges, industrial structures, towers and engineering applications where strength and weldability are important. Hot rolling produces plates, sheets and strips with specified mechanical properties.

Applicable IS:

IS 2062:2011

Standard title:

Hot Rolled Medium and High Tensile Structural Steel — Specification

Scope / applicability:

Hot rolled structural steel products covered by the grades, qualities, dimensions and delivery conditions specified by the standard.

BIS certification:

Mandatory where covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark.

Applicable QCO:

Steel and Steel Products (Quality Control) Order, 2024, as amended.

Current status — August 2026:

Listed within the BIS compulsory steel-certification framework where applicable.

Product Manual:

Applicable BIS Product Manual for IS 2062.

SIT:

Covers chemical composition, mechanical properties, impact requirements where applicable, dimensions and other product-specific controls.

Required testing:

- Chemical composition
- Yield strength
- Tensile strength
- Elongation
- Bend test
- Impact test where applicable
- Thickness
- Dimensions
- Surface quality
- Ultrasonic testing where specified
- Weldability-related requirements
- Other grade-specific tests

Manufacturing / quality requirements:

Steelmaking, casting, rolling and heat-treatment processes must be controlled to achieve the specified grade and quality.

Documents / certification requirements:

- BIS application
- Manufacturing-process details
- Grade/quality details
- Quality-control plan
- Testing capability
- Calibration records
- Chemical-analysis records
- Mechanical-test records
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark together with grade, quality, size and manufacturer/licence information as prescribed.

Important exclusions:

Special structural steels and products governed by newer dedicated standards must not automatically be classified under IS 2062.

2026 special note:

The AI should recognize "structural steel" as a broad consumer/industry phrase and ask for the grade or product form where necessary.

Official BIS source:

BIS — Scheme-I Steel and Iron Products

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/scheme-i-mark-scheme/>

CHUNK 13

Product:

High Strength Low Alloy Steel Plates and Strips

Description:

High-strength low-alloy steels are engineered to provide improved strength and toughness compared with ordinary carbon steel. They are used in structural, transportation, pressure and heavy-engineering applications.

Applicable IS:

Product-specific BIS classification must be verified according to grade, thickness and intended application.

Standard / classification:

Applicable IS depends on the particular HSLA product and grade.

BIS certification:

Mandatory where the exact product falls under the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark where applicable.

Applicable QCO:

Steel and Steel Products (Quality Control) Order, 2024, as amended.

Current status — August 2026:

Scope and exact standard require product-specific verification.

Product Manual:

To be identified after confirming the exact IS standard.

SIT:

Depends on the applicable product standard and grade.

Required testing:

- Chemical composition
- Yield strength
- Tensile strength
- Elongation
- Impact properties where specified
- Hardness where specified
- Thickness
- Dimensions
- Surface quality
- Bend test where specified
- Other grade-specific tests

Manufacturing / quality requirements:

Controlled alloying, rolling and heat treatment are important for achieving the specified high-strength properties.

Documents / certification requirements:

- BIS application

- Exact product standard
- Manufacturing-process details
- Grade range
- Quality-control plan
- Testing capability
- Calibration
- Test records
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark only where the product is covered by the applicable compulsory certification requirements.

Important exclusions:

Ordinary structural steel, pressure-vessel steel and automotive-grade steel may have separate standards.

2026 special note:

This product should remain a scope-verified category in the AI database until the user's exact steel grade and application are identified.

Official BIS source:

BIS — Products Under Compulsory Certification

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/>

Product:

Steel Plates for Pressure Vessels

Description:

Pressure-vessel steel plates are manufactured for equipment that contains fluids or gases under pressure. Their strength, toughness, weldability and resistance to service conditions are critical for safety.

Applicable IS:

Product-specific standard depends on the pressure-vessel steel grade and application.

Standard / classification:

The exact IS standard must be selected according to the intended pressure-vessel application and steel grade.

BIS certification:

Mandatory where the specific product is covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark where applicable.

Applicable QCO:

Steel and Steel Products (Quality Control) Order, 2024, as amended.

Current status — August 2026:

Scope verification required for exact standard and grade.

Product Manual:

To be identified after confirming the exact product classification.

SIT:

Depends on the applicable pressure-vessel steel standard.

Required testing:

- Chemical composition
- Tensile strength
- Yield strength
- Elongation
- Impact toughness where specified
- Bend test
- Thickness
- Dimensions
- Surface quality
- Ultrasonic testing where specified
- Heat-treatment verification
- Other grade-specific requirements

Manufacturing / quality requirements:

Steelmaking, rolling and heat treatment must be controlled and documented.
Traceability of heat/batch and test results is particularly important.

Documents / certification requirements:

- BIS application
- Product standard
- Manufacturing-process documentation
- Heat-treatment records
- Chemical-analysis reports
- Mechanical-test reports

- Calibration records
- Traceability records
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark and prescribed grade, heat/batch and manufacturer identification.

Important exclusions:

Finished pressure vessels are different from the steel plates used to manufacture them.

2026 special note:

The AI should distinguish the ****steel material**** from the ****finished pressure vessel**** when answering compliance questions.

Official BIS source:

BIS — Scheme-I Steel and Iron Products

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/scheme-i-mark-scheme/>

CHUNK 15

Product:

Steel Castings for General Engineering Purposes

Description:

Steel castings are components produced by pouring molten steel into moulds to obtain a required shape. They are used in machinery, engineering equipment, transport and industrial applications.

Applicable IS:

Product-specific BIS standard depends on the casting type, grade and application.

Standard / classification:

Exact IS classification must be established according to the casting grade and intended application.

BIS certification:

Mandatory where covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark where applicable.

Applicable QCO:

Steel and Steel Products (Quality Control) Order framework.

Current status — August 2026:

Product-specific scope verification required.

Product Manual:

To be identified for the exact steel-casting standard.

SIT:

Depends on the applicable grade and product standard.

Required testing:

- Chemical composition
- Tensile strength
- Yield strength where specified
- Elongation
- Hardness where specified
- Impact test where specified
- Dimensional inspection
- Surface examination
- Internal-defect examination where specified
- Heat-treatment verification
- Other grade-specific tests

Manufacturing / quality requirements:

Foundry controls, melt chemistry, moulding, casting, heat treatment and finishing processes must be controlled.

Documents / certification requirements:

- BIS application
- Applicable Indian Standard
- Foundry process details
- Heat/batch records
- Chemical-analysis records
- Mechanical-test reports
- Calibration records
- Inspection records
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark with applicable identification and traceability information.

Important exclusions:

Iron castings and non-ferrous castings belong to different material categories.

2026 special note:

A user saying "steel casting" should not receive a single universal IS number. The assistant should identify the grade/application first.

Official BIS source:

BIS — Products Under Compulsory Certification

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/>

CHUNK 16

Product:

Steel Forgings

Description:

Steel forgings are manufactured by plastically deforming heated steel under compressive forces to produce strong, durable components. They are widely used in automotive, railway, power and heavy-engineering applications.

Applicable IS:

Product-specific BIS standard depends on the type, grade and intended application.

Standard / classification:

Exact IS classification requires verification based on the forging type and application.

BIS certification:

Mandatory where covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark where applicable.

Applicable QCO:

Steel and Steel Products (Quality Control) Order framework.

Current status — August 2026:

Scope verification required.

Product Manual:

To be identified after determining the applicable forging standard.

SIT:

Depends on the exact forging product and grade.

Required testing:

- Chemical composition
- Tensile strength
- Yield strength where specified
- Elongation
- Hardness
- Impact properties where specified
- Dimensional inspection
- Surface quality

- Heat-treatment verification
- Grain/structure requirements where specified
- Non-destructive testing where required

Manufacturing / quality requirements:

Controlled forging temperature, reduction, heat treatment, machining and inspection processes are required.

Documents / certification requirements:

- BIS application
- Applicable standard
- Manufacturing-process flow
- Heat-treatment records
- Chemical-analysis reports
- Mechanical-test reports
- NDT records where applicable
- Calibration
- Traceability
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark with required manufacturer, grade and traceability identification.

Important exclusions:

Forgings for aircraft, pressure vessels or other specialized applications may have separate standards.

2026 special note:

The AI should distinguish ****forged steel material/components**** from cast steel products.

Official BIS source:

BIS — Scheme-I Steel and Iron Products

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/scheme-i-mark-scheme/>

CHUNK 17

Product:

Steel Tubes for Water Wells

Description:

Steel water-well tubes are tubular steel products used in groundwater extraction and well construction. They require adequate strength, dimensional accuracy and resistance to service conditions.

Applicable IS:

IS 4270:2001

Standard title:

Steel Tubes for Water Wells — Specification

Scope / applicability:

Steel tubes intended for water-well applications within the grades, dimensions and manufacturing requirements specified by the standard.

BIS certification:

Mandatory where covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark.

Applicable QCO:

Steel and Steel Products (Quality Control) Order, 2024, as amended.

Current status — August 2026:

Listed in BIS steel/iron compulsory-certification material where applicable.

Product Manual:

Applicable BIS Product Manual for IS 4270.

SIT:

Product-specific testing covering dimensions, chemical composition, mechanical properties and pressure/service requirements as specified.

Required testing:

- Chemical composition
- Tensile strength
- Yield strength where applicable
- Elongation
- Hydrostatic test where specified
- Dimensions
- Wall thickness
- Straightness
- Surface condition
- Thread/joint requirements where applicable

- Other standard-specific tests

Manufacturing / quality requirements:

Controlled tube manufacturing, welding or forming where applicable, finishing and dimensional inspection are required.

Documents / certification requirements:

- BIS application
- Manufacturing-process details
- Grade/product range
- QC plan
- Testing capability
- Calibration
- Routine test records
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark with size, grade, manufacturer and licence identification as prescribed.

Important exclusions:

Oil and gas line pipes, structural tubes and precision tubes may be governed by separate standards.

2026 special note:

The AI should ask whether a tube is intended for **water wells, structural use, plumbing, oil/gas or another application** before selecting an IS.

Official BIS source:

BIS — Products Under Compulsory Certification

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/>

CHUNK 18

Product:

Steel Tubes for Structural Purposes

Description:

Structural steel tubes are hollow steel sections used in buildings, towers, frames, supports and engineering structures. Their dimensional accuracy and mechanical strength are important for structural performance.

Applicable IS:

Product-specific standard depends on the tube type and manufacturing method.

Standard / classification:

Exact BIS classification should be selected according to the structural tube category.

BIS certification:

Mandatory where covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark where applicable.

Applicable QCO:

Steel and Steel Products (Quality Control) Order framework.

Current status — August 2026:

Scope verification required for exact standard.

Product Manual:

To be identified after confirming the applicable tube standard.

SIT:

Depends on the exact tube type.

Required testing:

- Chemical composition
- Tensile strength
- Yield strength
- Elongation
- Bend test
- Flattening test where applicable
- Hydrostatic test where specified
- Dimensions
- Wall thickness
- Straightness
- Surface condition
- Weld quality where applicable

Manufacturing / quality requirements:

Controlled tube forming, welding, heat treatment and finishing processes are required according to the applicable product standard.

Documents / certification requirements:

- BIS application

- Applicable IS
- Manufacturing-process details
- Welding/process controls where applicable
- QC plan
- Testing facilities
- Calibration
- Test records
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark with required size, grade and manufacturer identification.

Important exclusions:

Structural hollow sections covered by dedicated standards should not automatically be classified under a generic tube category.

2026 special note:

For construction queries, the AI should request the tube shape and standard/grade where the user only says "MS structural pipe."

Official BIS source:

BIS — Scheme-I Steel and Iron Products

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/scheme-i-mark-scheme/>

Product:

Steel Pipes for Potable Water Supply

Description:

Steel pipes used for drinking-water supply transport potable water through distribution systems. Their material, internal condition, corrosion protection and dimensional properties are important because the product comes into contact with drinking water.

Applicable IS:

Product-specific BIS standard must be verified according to pipe type, coating and application.

Standard / classification:

The exact IS depends on the pipe construction, manufacturing process and intended potable-water application.

BIS certification:

Mandatory where covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark where applicable.

Applicable QCO:

Steel and Steel Products (Quality Control) Order framework, subject to current scope.

Current status — August 2026:

Scope verification required for exact product classification.

Product Manual:

To be identified after confirming the applicable pipe standard.

SIT:

Depends on the applicable pipe standard and coating system.

Required testing:

- Chemical composition
- Tensile properties
- Yield strength
- Elongation
- Hydrostatic pressure
- Dimensions
- Wall thickness
- Weld quality where applicable
- Coating characteristics
- Surface condition
- Corrosion-related requirements where specified
- Potable-water contact requirements where applicable

Manufacturing / quality requirements:

Pipe forming, welding, heat treatment, coating and finishing must be controlled according to the applicable standard.

Documents / certification requirements:

- BIS application
- Applicable IS
- Manufacturing-process details
- Coating details

- QC plan
- Testing facilities
- Calibration
- Test records
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark with required size, grade, manufacturer and licence information.

Important exclusions:

Plastic potable-water pipes and specialized lined/coated systems may fall under different standards.

2026 special note:

The AI must not assume that every "water pipe" is suitable for potable water. Intended use must be identified.

Official BIS source:

BIS — Products Under Compulsory Certification

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/>

CHUNK 20

Product:

Steel Pipes for Oil and Gas Transportation

Description:

Steel line pipes are used to transport oil, natural gas and other fluids over long distances and within industrial facilities. They must meet demanding requirements for strength, toughness, weld integrity and dimensional consistency.

Applicable IS:

Product-specific BIS standard depends on the line-pipe type and application.

Standard / classification:

The applicable Indian Standard must be identified according to pipe grade, manufacturing method, service and intended use.

BIS certification:

Mandatory where the exact product falls within the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark where applicable.

Applicable QCO:

Steel and Steel Products (Quality Control) Order, 2024, as amended.

Current status — August 2026:

Scope verification required for the exact line-pipe category.

Product Manual:

To be identified after confirming the applicable standard.

SIT:

Product-specific testing according to the applicable line-pipe standard.

Required testing:

- Chemical composition
- Tensile strength
- Yield strength
- Elongation
- Hydrostatic pressure
- Flattening/bend requirements where specified
- Impact toughness where specified
- Weld quality
- Dimensional tolerances
- Wall thickness
- Surface condition
- Non-destructive examination where specified
- Other grade/service-specific tests

Manufacturing / quality requirements:

Steelmaking, plate/strip production, forming, welding, heat treatment, inspection and hydrostatic testing require controlled processes and traceability.

Documents / certification requirements:

- BIS application
- Applicable Indian Standard
- Manufacturing process flow
- Grade range
- Welding procedure/process controls
- Heat-treatment records
- NDT records
- Mechanical-test records
- Calibration

- Traceability records
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark together with pipe grade, dimensions, heat/batch and manufacturer/licence identification as prescribed.

Important exclusions:

Pipes for water supply, structural applications and high-temperature steam service can have different standards and requirements.

2026 special note:

For oil and gas products, the AI should not give a generic "BIS certified" answer. It should identify the exact pipe standard, grade and service before determining compliance.

Official BIS source:

BIS — Scheme-I Steel and Iron Products

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/scheme-i-mark-scheme/>

CHUNK 21

Product:

Steel Pipes for Boiler and Heat Exchanger Applications

Description:

These are steel tubes and pipes designed for boilers, heat exchangers and other high-temperature service environments. They must withstand pressure, temperature and thermal cycling while maintaining adequate mechanical integrity.

Applicable IS:

Product-specific BIS standard depends on the exact tube type, grade and service.

Standard / classification:

The applicable IS must be identified according to whether the product is intended for boilers, heat exchangers, superheaters or another specialized application.

BIS certification:

Mandatory where the exact product is covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark where applicable.

Applicable QCO:

Steel and Steel Products (Quality Control) Order, 2024, as amended.

Current status — August 2026:

Exact standard and scope require product-specific verification.

Product Manual:

To be identified after confirming the applicable IS.

SIT:

Depends on the applicable boiler/heat-exchanger tube standard.

Required testing:

- Chemical composition
- Tensile strength
- Yield strength where specified
- Elongation
- Hydrostatic test
- Flattening test
- Flaring test where specified
- Dimensions
- Wall thickness
- Surface condition
- Heat-treatment verification
- Non-destructive testing where specified
- Other grade-specific requirements

Manufacturing / quality requirements:

Controlled tube manufacturing, heat treatment, dimensional control and inspection are required for high-temperature and pressure service.

Documents / certification requirements:

- BIS application
- Applicable Indian Standard
- Manufacturing-process details
- Grade range
- Heat-treatment records
- Mechanical-test reports
- NDT records where applicable
- Calibration records
- Traceability
- Product Manual/SIT

- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark with applicable grade, size, heat/batch and manufacturer identification.

Important exclusions:

Ordinary structural tubes and general-purpose pipes should not be classified as boiler tubes without confirming their service/application.

2026 special note:

For safety-critical applications, the AI should identify the service temperature, pressure and intended equipment before recommending a standard.

Official BIS source:

BIS — Products Under Compulsory Certification

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/>

CHUNK 22

Product:

Seamless Carbon Steel Tubes for High-Temperature Service

Description:

Seamless carbon steel tubes are manufactured without a welded seam and are used in applications involving elevated temperature, pressure or demanding mechanical conditions. They are commonly used in industrial equipment and process systems.

Applicable IS:

Product-specific standard depends on grade, dimensions and intended service.

Standard / classification:

Exact IS must be selected based on the application and material grade.

BIS certification:

Mandatory where covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark where applicable.

Applicable QCO:

Steel and Steel Products (Quality Control) Order framework.

Current status — August 2026:

Scope verification required.

Product Manual:

To be identified after confirming the applicable standard.

SIT:

Product-specific testing.

Required testing:

- Chemical composition
- Tensile strength
- Yield strength
- Elongation
- Hydrostatic pressure

- Flattening test
- Flaring test where applicable
- Hardness
- Dimensions
- Wall thickness
- Surface quality
- Heat treatment
- Non-destructive testing where specified

Manufacturing / quality requirements:

Seamless tube manufacturing, heat treatment, finishing and inspection must be controlled to achieve the required mechanical and dimensional characteristics.

Documents / certification requirements:

- BIS application
- Applicable IS
- Manufacturing-process flow
- Heat-treatment records
- Chemical analysis
- Mechanical tests
- NDT records
- Calibration
- Traceability
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark with grade, dimensions and traceability information.

Important exclusions:

Welded tubes and specialized alloy tubes may be governed by different standards.

2026 special note:

The AI should not assume that "seamless pipe" automatically means one particular BIS standard.

Official BIS source:

BIS — Scheme-I Steel and Iron Products

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/scheme-i-mark-scheme/>

CHUNK 23

Product:

Stainless Steel Plate, Sheet and Strip

Description:

Stainless steel is a corrosion-resistant alloy steel containing chromium and, depending on grade, nickel, molybdenum and other alloying elements. Sheets, plates and strips are widely used in food processing, chemical equipment, construction, architecture and engineering.

Applicable IS:

Product-specific standard depends on the stainless-steel grade and product form.

Standard / classification:

The exact IS must be selected based on grade, finish, thickness and application.

BIS certification:

Mandatory where the exact product is covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark where applicable.

Applicable QCO:

Steel and Steel Products (Quality Control) Order, 2024, as amended.

Current status — August 2026:

Exact IS and compulsory scope require product-specific verification.

Product Manual:

To be identified for the exact stainless-steel standard.

SIT:

Depends on the grade and product standard.

Required testing:

- Chemical composition
- Chromium content
- Nickel content where applicable
- Molybdenum where applicable
- Tensile strength
- Yield strength
- Elongation
- Hardness where specified
- Corrosion-related requirements
- Dimensions

- Thickness
- Surface finish
- Bend test where specified

Manufacturing / quality requirements:

Alloy control, melting, rolling, annealing, pickling and finishing processes must be controlled according to the grade.

Documents / certification requirements:

- BIS application
- Applicable IS
- Grade/product range
- Manufacturing process
- Heat/batch records
- Chemical analysis
- Mechanical testing
- Surface inspection
- Calibration
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark with grade, size and manufacturer/licence information as prescribed.

Important exclusions:

Stainless steel utensils and finished stainless-steel consumer products may have separate standards.

2026 special note:

The AI should distinguish stainless-steel raw material from finished utensils, kitchen equipment and fabricated components.

Official BIS source:

BIS — Products Under Compulsory Certification

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/>

CHUNK 24

Product:

Stainless Steel Wire

Description:

Stainless steel wire is corrosion-resistant wire used in springs, mesh, fasteners, medical applications, engineering products and general fabrication. Different grades provide different combinations of strength, corrosion resistance and ductility.

Applicable IS:

Product-specific BIS standard depends on wire grade, finish and intended application.

Standard / classification:

Exact IS must be selected according to the wire category and application.

BIS certification:

Mandatory where covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark where applicable.

Applicable QCO:

Steel and Steel Products (Quality Control) Order framework.

Current status — August 2026:

Scope verification required.

Product Manual:

To be identified after confirming the applicable wire standard.

SIT:

Product-specific chemical, mechanical and dimensional testing.

Required testing:

- Chemical composition
- Tensile strength
- Elongation
- Hardness where specified
- Diameter
- Dimensional tolerance
- Surface quality
- Corrosion-related requirements where applicable
- Coating requirements where applicable
- Other grade-specific tests

Manufacturing / quality requirements:

Controlled drawing, heat treatment, finishing and surface treatment are required according to the grade and intended application.

Documents / certification requirements:

- BIS application
- Applicable standard
- Grade details
- Manufacturing process
- Test facilities
- Calibration
- Test records
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark where applicable, together with grade and manufacturer/licence information.

Important exclusions:

Spring wire, welding wire and specialized medical/industrial wire can fall under separate standards.

2026 special note:

The AI should identify the intended use before selecting the applicable stainless-steel wire standard.

Official BIS source:

BIS — Scheme-I Steel and Iron Products

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/scheme-i-mark-scheme/>

CHUNK 25

Product:

Stainless Steel Bars and Rods

Description:

Stainless steel bars and rods are solid stainless-steel products used for shafts, fasteners, machine components, fabrication and structural applications. Their grade determines corrosion resistance and mechanical performance.

Applicable IS:

Product-specific BIS standard depends on the stainless-steel grade and bar/rod form.

Standard / classification:

Exact IS must be identified according to grade, dimensions and intended use.

BIS certification:

Mandatory where covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark where applicable.

Applicable QCO:

Steel and Steel Products (Quality Control) Order, 2024, as amended.

Current status — August 2026:

Product-specific scope verification required.

Product Manual:

To be identified for the exact stainless-steel bar/rod standard.

SIT:

Product-specific chemical, mechanical and dimensional testing.

Required testing:

- Chemical composition
- Tensile strength
- Yield strength
- Elongation
- Hardness where specified
- Bend test where specified
- Diameter/section dimensions
- Surface condition
- Straightness
- Corrosion-related requirements where specified

Manufacturing / quality requirements:

Controlled melting, hot/cold working, heat treatment and finishing processes are required according to grade.

Documents / certification requirements:

- BIS application
- Applicable IS
- Grade/product range
- Manufacturing process
- Heat-treatment records
- Chemical analysis

- Mechanical-test records
- Calibration
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark with grade, dimensions and manufacturer/licence identification.

Important exclusions:

Special stainless-steel products for aerospace, medical or pressure applications may be governed by separate standards.

2026 special note:

The AI should not treat "stainless rod" as a sufficient classification because grade and application can materially change the applicable standard.

Official BIS source:

BIS — Products Under Compulsory Certification

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/>

CHUNK 26

Product:

High Carbon Steel Wire

Description:

High carbon steel wire contains a relatively high carbon level and is processed to achieve high strength and hardness. It is used in springs, ropes, tyres, mechanical components and other engineering applications.

Applicable IS:

Product-specific BIS standard depends on the intended application.

Standard / classification:

Exact IS must be selected based on the wire type, grade and end use.

BIS certification:

Mandatory where covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark where applicable.

Applicable QCO:

Steel and Steel Products (Quality Control) Order framework.

Current status — August 2026:

Scope verification required.

Product Manual:

To be identified after confirming the exact wire standard.

SIT:

Product-specific testing.

Required testing:

- Chemical composition

- Tensile strength
- Elongation
- Diameter
- Dimensional tolerance
- Surface condition
- Hardness where specified
- Torsion test where applicable
- Wrap/bend test where applicable
- Other application-specific tests

Manufacturing / quality requirements:

Controlled wire rod quality, drawing, heat treatment and finishing processes are important.

Documents / certification requirements:

- BIS application
- Applicable IS
- Grade details
- Manufacturing process
- Test facilities
- Calibration
- Test records
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark where applicable with grade and manufacturer/licence identification.

Important exclusions:

Spring wire, tyre cord and prestressing wire have specialized requirements and should not automatically be mapped to a generic high-carbon-wire category.

2026 special note:

Application is essential for this product because several specialized high-carbon steel wires exist.

Official BIS source:

BIS — Scheme-I Steel and Iron Products

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/scheme-i-mark-scheme/>

CHUNK 27

Product:

Steel Wire Ropes

Description:

Steel wire ropes consist of multiple steel wires laid together to form strands and ropes. They are used for lifting, cranes, elevators, mining, material handling and other applications where high tensile strength and controlled flexibility are required.

Applicable IS:

Product-specific BIS standard depends on rope construction and intended application.

Standard / classification:

The exact IS must be selected according to rope construction, grade, diameter and service.

BIS certification:

Mandatory where the exact product is covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark where applicable.

Applicable QCO:

Steel and Steel Products (Quality Control) Order framework.

Current status — August 2026:

Exact compulsory scope requires verification.

Product Manual:

To be identified for the applicable steel wire-rope standard.

SIT:

Product-specific testing and inspection.

Required testing:

- Wire chemical composition
- Wire tensile strength
- Rope breaking load
- Rope diameter
- Lay characteristics
- Construction
- Core characteristics
- Lubrication where specified
- Surface condition

- Dimensional requirements
- Elongation where specified
- Other rope-specific tests

Manufacturing / quality requirements:

Wire drawing, stranding, closing, lubrication and inspection processes must be controlled and documented.

Documents / certification requirements:

- BIS application
- Applicable IS
- Rope construction details
- Wire-grade information
- Manufacturing-process documentation
- Breaking-load test records
- Calibration
- Traceability
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark where applicable with rope size, construction, grade and manufacturer identification.

Important exclusions:

Elevator ropes, aircraft cables and specialized high-performance ropes may have separate standards.

2026 special note:

The AI should ask about the application when a user asks simply "steel wire rope BIS standard."

Official BIS source:

BIS — Products Under Compulsory Certification

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/>

CHUNK 28

Product:

Steel Nails

Description:

Steel nails are fastening products used in construction, woodworking, packaging and general fabrication. Their dimensions, material and mechanical performance determine their suitability for different applications.

Applicable IS:

Product-specific BIS standard depends on nail type and intended application.

Standard / classification:

Exact IS must be identified according to nail design, material and use.

BIS certification:

Mandatory where covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark where applicable.

Applicable QCO:

Steel and Steel Products (Quality Control) Order framework.

Current status — August 2026:

Scope verification required for the exact nail category.

Product Manual:

To be identified after confirming the applicable standard.

SIT:

Product-specific testing.

Required testing:

- Material/chemical composition
- Dimensions
- Diameter
- Length
- Head dimensions
- Point geometry
- Tensile/shear-related properties where specified
- Bending performance
- Surface condition
- Coating where applicable

Manufacturing / quality requirements:

Wire preparation, nail forming, heading, pointing and surface finishing should be controlled.

Documents / certification requirements:

- BIS application
- Applicable IS
- Manufacturing-process details
- Product dimensions
- Testing facilities
- Calibration
- Test records
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark where applicable with prescribed product identification.

Important exclusions:

Concrete nails, roofing nails, specialized fasteners and coated nails may be covered by different specifications.

2026 special note:

The assistant should ask for the nail type/application before selecting an IS standard.

Official BIS source:

BIS — Scheme-I Steel and Iron Products

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/scheme-i-mark-scheme/>

Product:

Steel Bolts and Screws

Description:

Steel bolts and screws are mechanical fasteners used to join structural, automotive, machinery and fabricated components. Their strength class, dimensions, thread geometry and material properties are critical for safe connections.

Applicable IS:

Product-specific BIS standards apply depending on fastener type, dimensions, thread and property class.

Standard / classification:

The exact IS must be selected according to whether the product is a structural bolt, machine screw, hexagonal bolt or another fastener.

BIS certification:

Mandatory where covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark where applicable.

Applicable QCO:

Steel and Steel Products (Quality Control) Order framework.

Current status — August 2026:

Exact scope requires product-specific verification.

Product Manual:

To be identified according to the exact fastener standard.

SIT:

Product-specific mechanical, dimensional and material testing.

Required testing:

- Chemical composition
- Tensile strength
- Proof load where applicable
- Yield strength where specified
- Hardness
- Thread dimensions
- Pitch
- Head dimensions
- Surface finish
- Coating
- Decarburization where applicable
- Dimensional tolerances

Manufacturing / quality requirements:

Controlled heading, threading, heat treatment and coating processes are important for achieving specified fastener properties.

Documents / certification requirements:

- BIS application
- Applicable IS
- Grade/property class
- Manufacturing process
- Heat-treatment records
- Test facilities

- Calibration
- Test records
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark where applicable, together with manufacturer and grade/property-class identification.

Important exclusions:

Stainless-steel fasteners and specialized aerospace/automotive fasteners may be governed by separate standards.

2026 special note:

The AI should never select an IS only from the word "bolt." It should identify the fastener type, dimensions, property class and application.

Official BIS source:

BIS — Products Under Compulsory Certification

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/>

CHUNK 30

Product:

Steel Nuts

Description:

Steel nuts are threaded fasteners used with bolts and screws to create detachable mechanical joints. Their thread accuracy, dimensions, material strength and compatibility with the mating bolt are essential for reliable fastening.

Applicable IS:

Product-specific BIS standard depends on nut type, dimensions, thread and property class.

Standard / classification:

Exact IS must be selected according to the nut design and intended application.

BIS certification:

Mandatory where covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark where applicable.

Applicable QCO:

Steel and Steel Products (Quality Control) Order framework.

Current status — August 2026:

Exact compulsory scope requires product-specific verification.

Product Manual:

To be identified for the applicable nut standard.

SIT:

Product-specific mechanical, dimensional and material testing.

Required testing:

- Chemical composition
- Proof load
- Hardness
- Thread dimensions
- Thread pitch
- Across-flats dimensions
- Height
- Dimensional tolerances
- Surface condition
- Coating where applicable
- Decarburization where applicable
- Other property-class requirements

Manufacturing / quality requirements:

Controlled forging/heading, threading, heat treatment and surface finishing processes are required according to the applicable grade and property class.

Documents / certification requirements:

- BIS application
- Applicable IS
- Nut type/property class
- Manufacturing-process details
- Heat-treatment records
- Testing facilities
- Calibration
- Test records
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark where applicable with manufacturer and property-class identification.

Important exclusions:

Special nuts for aerospace, automotive, pressure equipment or stainless-steel applications may have separate standards.

2026 special note:

The AI should pair nut and bolt requirements where possible because compatibility of thread, dimensions and property class is essential for a safe fastening system.

Official BIS source:

BIS — Products Under Compulsory Certification

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/>

CHUNK 31

Product:

Steel Rivets

Description:

Steel rivets are permanent mechanical fasteners used to join sheets, plates and structural components. They are widely used in fabrication, engineering and certain structural applications where a permanent joint is required.

Applicable IS:

Product-specific BIS standard depends on rivet type, dimensions, material and application.

Standard / classification:

The exact IS must be selected according to the rivet design, head type, dimensions and steel grade.

BIS certification:

Mandatory where the exact product is covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark where applicable.

Applicable QCO:

Steel and Steel Products (Quality Control) Order framework.

Current status — August 2026:

Exact compulsory scope requires product-specific verification.

Product Manual:

To be identified after confirming the applicable rivet standard.

SIT:

Product-specific dimensional, material and mechanical testing.

Required testing:

- Chemical composition
- Tensile/shear properties where specified
- Dimensions
- Rivet diameter
- Length
- Head dimensions

- Straightness
- Surface condition
- Material condition
- Hardness where specified
- Other standard-specific requirements

Manufacturing / quality requirements:

Controlled heading, forming, heat treatment where applicable and dimensional finishing are required to maintain consistency.

Documents / certification requirements:

- BIS application
- Applicable IS
- Manufacturing-process details
- Grade/material details
- Testing capability
- Calibration records
- Routine test records
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark where applicable, together with prescribed product identification.

Important exclusions:

Aluminium, copper and specialized aerospace rivets are governed by different material/product requirements.

2026 special note:

The AI should distinguish rivets from bolts and screws because they create permanent rather than normally detachable joints.

Official BIS source:

BIS — Products Under Compulsory Certification

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/>

CHUNK 32

Product:

Steel Washers

Description:

Steel washers are flat or specially shaped components placed under nuts, bolts or other fasteners to distribute load and protect the mating surface. They are widely used in construction, machinery and fabrication.

Applicable IS:

Product-specific BIS standard depends on washer type, dimensions and application.

Standard / classification:

The exact IS should be identified according to washer type and dimensional series.

BIS certification:

Mandatory where covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark where applicable.

Applicable QCO:

Steel and Steel Products (Quality Control) Order framework.

Current status — August 2026:

Exact compulsory scope requires product-specific verification.

Product Manual:

To be identified for the applicable washer standard.

SIT:

Dimensional, material and mechanical testing as specified.

Required testing:

- Chemical composition
- Dimensions
- Inner diameter
- Outer diameter
- Thickness
- Flatness
- Hardness where specified
- Material properties
- Surface condition
- Coating where applicable
- Other dimensional requirements

Manufacturing / quality requirements:

Controlled blanking, stamping, forming, heat treatment and surface finishing processes where applicable.

Documents / certification requirements:

- BIS application
- Applicable IS
- Product dimensions
- Material/grade details
- QC procedures
- Testing capability
- Calibration
- Test records
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark where applicable with prescribed product identification.

Important exclusions:

Spring washers, lock washers and specialized washers may have dedicated standards.

2026 special note:

The AI should ask whether the user means a plain washer, spring washer, lock washer or another type before selecting a standard.

Official BIS source:

BIS — Scheme-I Steel and Iron Products

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/scheme-i-mark-scheme/>

CHUNK 33

Product:

Steel Chains

Description:

Steel chains are assemblies of interconnected metal links used for lifting, conveying, anchoring, transmission and material handling. Their strength and dimensional consistency are essential for safe operation.

Applicable IS:

Product-specific BIS standard depends on chain type and application.

Standard / classification:

Exact IS must be selected according to whether the chain is for lifting, conveyor service, transmission, anchoring or another purpose.

BIS certification:

Mandatory where covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark where applicable.

Applicable QCO:

Steel and Steel Products (Quality Control) Order framework.

Current status — August 2026:

Exact compulsory scope requires verification.

Product Manual:

To be identified according to the applicable chain standard.

SIT:

Product-specific mechanical, dimensional and material testing.

Required testing:

- Chemical composition
- Link dimensions
- Pitch
- Diameter
- Proof load
- Breaking load
- Elongation
- Surface condition
- Hardness where specified
- Heat-treatment verification where applicable
- Other chain-specific tests

Manufacturing / quality requirements:

Controlled wire/rod preparation, link forming, welding, heat treatment and finishing are required according to chain type.

Documents / certification requirements:

- BIS application
- Applicable IS
- Chain type
- Grade/material details
- Manufacturing-process documentation

- Load-test records
- Calibration
- Traceability
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark where applicable, with chain size, grade and manufacturer identification as prescribed.

Important exclusions:

Lifting chains, conveyor chains and transmission chains may be governed by different standards.

2026 special note:

The assistant should always identify the chain's intended use because safety requirements vary substantially between lifting and general-purpose applications.

Official BIS source:

BIS — Products Under Compulsory Certification

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/>

CHUNK 34

Product:

Steel Structural Sections — Angles

Description:

Steel angles are L-shaped structural sections used in trusses, frames, towers, supports and fabricated structures. They are commonly produced by hot rolling and are selected according to grade, dimensions and structural requirements.

Applicable IS:

Product-specific structural-steel standard depends on grade and section type.

Standard / classification:

The applicable IS should be identified according to whether the angle is equal or unequal and the applicable steel grade.

BIS certification:

Mandatory where covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark where applicable.

Applicable QCO:

Steel and Steel Products (Quality Control) Order framework.

Current status — August 2026:

Exact standard and compulsory scope require verification.

Product Manual:

To be identified according to the applicable structural-section standard.

SIT:

Chemical, mechanical and dimensional testing.

Required testing:

- Chemical composition
- Yield strength
- Tensile strength
- Elongation
- Bend test
- Leg dimensions
- Thickness
- Mass per metre
- Straightness
- Surface condition
- Dimensional tolerances

Manufacturing / quality requirements:

Controlled steelmaking and hot rolling are required to maintain section geometry and mechanical properties.

Documents / certification requirements:

- BIS application
- Applicable IS
- Grade/product range
- Manufacturing process
- QC plan
- Testing facilities
- Calibration
- Test records
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark with grade, size and manufacturer/licence identification.

Important exclusions:

Aluminium angles and cold-formed sections are different products and may have different standards.

2026 special note:

The AI should interpret "MS angle" as a possible structural-steel angle query but should verify grade and applicable standard.

Official BIS source:

BIS — Scheme-I Steel and Iron Products

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/scheme-i-mark-scheme/>

CHUNK 35

Product:

Steel Structural Sections — Channels

Description:

Steel channels are U- or C-shaped structural sections used in building frames, machinery supports, bridges and fabrication. Their dimensions and mechanical properties determine load-carrying performance.

Applicable IS:

Product-specific structural-section standard depends on grade and section type.

Standard / classification:

The exact IS should be selected according to channel dimensions, manufacturing method and steel grade.

BIS certification:

Mandatory where covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark where applicable.

Applicable QCO:

Steel and Steel Products (Quality Control) Order framework.

Current status — August 2026:

Exact standard and compulsory scope require verification.

Product Manual:

To be identified for the applicable channel standard.

SIT:

Chemical, mechanical and dimensional testing.

Required testing:

- Chemical composition
- Yield strength
- Tensile strength
- Elongation
- Bend test
- Web dimensions

- Flange dimensions
- Thickness
- Mass per metre
- Straightness
- Surface condition
- Dimensional tolerances

Manufacturing / quality requirements:

Controlled hot rolling, cooling and finishing processes are required for consistent section dimensions and properties.

Documents / certification requirements:

- BIS application
- Applicable IS
- Grade/product range
- Manufacturing process
- QC plan
- Test facilities
- Calibration
- Routine test records
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark with prescribed size, grade and manufacturer identification.

Important exclusions:

Cold-formed channels and specialized light-gauge sections may fall under different standards.

2026 special note:

The AI should distinguish hot-rolled structural channels from light-gauge cold-formed channels.

Official BIS source:

BIS — Products Under Compulsory Certification

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/>

CHUNK 36

Product:

Steel Structural Sections — Beams and Joists

Description:

Steel beams and joists are structural sections designed to carry bending and other loads in buildings, bridges, industrial structures and engineering works. They are commonly hot rolled and supplied in standardized sizes.

Applicable IS:

Product-specific standard depends on section type and steel grade.

Standard / classification:

Exact IS should be selected according to beam/joist section, dimensions and material grade.

BIS certification:

Mandatory where covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark where applicable.

Applicable QCO:

Steel and Steel Products (Quality Control) Order framework.

Current status — August 2026:

Exact standard and compulsory scope require verification.

Product Manual:

To be identified for the applicable structural-section standard.

SIT:

Chemical, mechanical and dimensional testing.

Required testing:

- Chemical composition
- Yield strength
- Tensile strength
- Elongation
- Bend test
- Section dimensions
- Web thickness
- Flange thickness
- Mass per metre
- Straightness
- Surface quality
- Dimensional tolerances

Manufacturing / quality requirements:

Controlled steelmaking, rolling and finishing processes are required to maintain section geometry and mechanical performance.

Documents / certification requirements:

- BIS application
- Applicable IS
- Grade/product range
- Manufacturing details
- Testing facilities
- Calibration
- Mechanical and chemical test records
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark with size, grade and manufacturer/licence identification.

Important exclusions:

Fabricated built-up beams and welded structural assemblies are different from rolled sections.

2026 special note:

The assistant should distinguish a rolled steel beam from a fabricated structural member made by welding plates or sections together.

Official BIS source:

BIS — Scheme-I Steel and Iron Products

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/scheme-i-mark-scheme/>

CHUNK 37

Product:

Steel Flats

Description:

Steel flats are rectangular solid steel sections used in fabrication, construction, machinery and general engineering. They are available in different widths and thicknesses and may be hot rolled or processed for specific applications.

Applicable IS:

Product-specific standard depends on the grade, dimensions and intended application.

Standard / classification:

Exact IS must be identified according to the steel grade and flat-bar category.

BIS certification:

Mandatory where covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark where applicable.

Applicable QCO:

Steel and Steel Products (Quality Control) Order framework.

Current status — August 2026:

Scope verification required.

Product Manual:

To be identified after confirming the applicable standard.

SIT:

Chemical, mechanical and dimensional testing.

Required testing:

- Chemical composition
- Yield strength where specified
- Tensile strength
- Elongation
- Bend test
- Width
- Thickness
- Mass per metre
- Straightness
- Surface condition
- Dimensional tolerances

Manufacturing / quality requirements:

Controlled rolling and finishing are necessary for maintaining dimensions and mechanical properties.

Documents / certification requirements:

- BIS application

- Applicable IS
- Grade/product range
- Manufacturing process
- QC plan
- Test facilities
- Calibration
- Test records
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark where applicable with grade, dimensions and manufacturer identification.

Important exclusions:

Special flat bars for springs, tools or other engineering applications may have dedicated standards.

2026 special note:

"MS flat" is a broad market term. The AI should identify grade and intended application before giving a definitive compliance answer.

Official BIS source:

BIS — Products Under Compulsory Certification

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/>

Product:

Steel Round Bars

Description:

Steel round bars are solid cylindrical steel sections used in shafts, fasteners, fabrication, machinery and structural applications. They are supplied in different grades and sizes depending on the required mechanical properties.

Applicable IS:

Product-specific standard depends on grade and intended application.

Standard / classification:

The exact IS should be identified according to the steel grade, manufacturing condition and end use.

BIS certification:

Mandatory where covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark where applicable.

Applicable QCO:

Steel and Steel Products (Quality Control) Order framework.

Current status — August 2026:

Exact compulsory scope requires verification.

Product Manual:

To be identified for the applicable round-bar standard.

SIT:

Chemical, mechanical and dimensional testing.

Required testing:

- Chemical composition
- Yield strength
- Tensile strength
- Elongation
- Hardness where specified
- Bend test where specified
- Diameter
- Straightness
- Mass per metre
- Surface condition
- Dimensional tolerances

Manufacturing / quality requirements:

Controlled steelmaking, rolling, heat treatment where applicable and finishing processes are required.

Documents / certification requirements:

- BIS application
- Applicable IS
- Grade/product range
- Manufacturing details
- Heat-treatment records where applicable
- Chemical analysis
- Mechanical tests
- Calibration

- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark with grade, size and manufacturer/licence information.

Important exclusions:

Reinforcement bars, tool steels and stainless-steel bars may be governed by separate standards.

2026 special note:

The assistant should not interpret "round bar" as a single BIS product because round bars exist across many steel grades and applications.

Official BIS source:

BIS — Scheme-I Steel and Iron Products

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/scheme-i-mark-scheme/>

CHUNK 39

Product:

Steel Square and Hexagonal Bars

Description:

Square and hexagonal steel bars are solid sections used in engineering, fabrication, fasteners and machine components. Their geometry allows specialized machining and fitting applications.

Applicable IS:

Product-specific standard depends on grade, section and intended application.

Standard / classification:

Exact IS must be identified according to steel grade and dimensional category.

BIS certification:

Mandatory where covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark where applicable.

Applicable QCO:

Steel and Steel Products (Quality Control) Order framework.

Current status — August 2026:

Scope verification required.

Product Manual:

To be identified after confirming the exact standard.

SIT:

Chemical, mechanical and dimensional testing.

Required testing:

- Chemical composition
- Tensile strength
- Yield strength where specified
- Elongation

- Hardness where specified
- Dimensions across flats
- Thickness/section size
- Straightness
- Surface quality
- Bend test where specified
- Dimensional tolerances

Manufacturing / quality requirements:

Controlled rolling/drawing, heat treatment where applicable and finishing processes must be maintained.

Documents / certification requirements:

- BIS application
- Applicable IS
- Grade/product range
- Manufacturing process
- Testing capability
- Calibration
- Test records
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark where applicable with grade, section and manufacturer/licence identification.

Important exclusions:

Cold-finished precision bars and specialized free-cutting steels may have separate standards.

2026 special note:

The AI should identify whether the user means a structural bar, machined bar, bright bar or specialized engineering bar.

Official BIS source:

BIS — Products Under Compulsory Certification

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/>

CHUNK 40

Product:

Bright Steel Bars

Description:

Bright steel bars are steel bars that undergo additional cold-finishing processes to obtain improved dimensional accuracy and surface finish. They are widely used in precision engineering, automotive and machine-component manufacturing.

Applicable IS:

Product-specific BIS standard depends on the bright-bar type and steel grade.

Standard / classification:

The exact IS must be identified according to the steel grade, section and finishing process.

BIS certification:

Mandatory where covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark where applicable.

Applicable QCO:

Steel and Steel Products (Quality Control) Order framework.

Current status — August 2026:

Exact compulsory scope requires verification.

Product Manual:

To be identified for the applicable bright-steel-bar standard.

SIT:

Chemical, mechanical, dimensional and surface-quality testing.

Required testing:

- Chemical composition
- Tensile strength
- Yield strength where specified
- Elongation
- Hardness
- Diameter/section dimensions
- Dimensional tolerance
- Straightness
- Surface roughness/finish where specified
- Surface defects
- Other grade-specific tests

Manufacturing / quality requirements:

Controlled hot-rolled input material, cold drawing/turning/peeling, straightening and finishing processes are required according to the product specification.

Documents / certification requirements:

- BIS application
- Applicable IS
- Grade/product range
- Manufacturing process
- Finishing process details
- Testing facilities
- Calibration
- Test records
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark where applicable with grade, dimensions and manufacturer/licence identification.

Important exclusions:

Precision bright bars, stainless bright bars and specialized engineering bars may fall under different standards.

2026 special note:

"Bright bar" describes a finished condition rather than one universal steel grade. The AI should identify the base material and exact product specification before giving compliance information.

Official BIS source:

BIS — Scheme-I Steel and Iron Products

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/scheme-i-mark-scheme/>

CHUNK 41

Product:

Steel Wire Rods

Description:

Steel wire rod is a hot-rolled semi-finished steel product supplied in coils and used as the starting material for manufacturing wires, fasteners, springs, nails and other steel products. Its chemical composition, dimensions and mechanical properties are important for downstream processing.

Applicable IS:

Product-specific BIS standard depends on the grade and intended application.

Standard / classification:

The applicable IS must be selected according to whether the wire rod is intended for general engineering, welding, prestressing, spring manufacture, tyre reinforcement or another specialized application.

BIS certification:

Mandatory where the exact product falls within the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark where applicable.

Applicable QCO:

Steel and Steel Products (Quality Control) Order, 2024, as amended.

Current status — August 2026:

Exact compulsory scope requires verification based on grade and intended application.

Product Manual:

To be identified for the applicable wire-rod standard.

SIT:

Product-specific chemical, mechanical, dimensional and surface testing.

Required testing:

- Chemical composition
- Carbon and alloying elements where applicable
- Tensile strength
- Yield strength where specified
- Elongation
- Diameter
- Dimensional tolerance
- Surface condition
- Decarburization where applicable
- Internal quality where specified
- Other grade-specific tests

Manufacturing / quality requirements:

Steelmaking, continuous casting, hot rolling, controlled cooling and coil finishing must be controlled according to the applicable grade.

Documents / certification requirements:

- BIS application
- Applicable IS

- Grade/product range
- Manufacturing process
- Heat/batch traceability
- Chemical-analysis records
- Mechanical-test records
- Calibration records
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark where applicable with grade, size, heat/batch and manufacturer/licence information.

Important exclusions:

Special wire rods for welding electrodes, prestressing, springs and tyre reinforcement may have dedicated standards.

2026 special note:

The AI should treat "wire rod" as an intermediate steel product and identify its intended downstream use before assigning a definitive standard.

Official BIS source:

BIS — Products Under Compulsory Certification

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/>

CHUNK 42

Product:

Prestressing Steel Wires

Description:

Prestressing steel wires are high-strength steel wires used to introduce compressive stresses into concrete structures. They are critical components in prestressed concrete bridges, buildings, sleepers and other infrastructure.

Applicable IS:

IS 1785 (Part 1):1983 and applicable parts/amendments depending on the product.

Standard title:

Plain and Hard-Drawn Steel Wire for Prestressed Concrete — Specification

Scope / applicability:

High-strength steel wire intended for prestressed concrete applications within the specified dimensions, grades and mechanical requirements.

BIS certification:

Mandatory where covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark.

Applicable QCO:

Steel and Steel Products (Quality Control) Order, 2024, as amended.

Current status — August 2026:

Covered by compulsory certification where the product falls within the applicable QCO scope.

Product Manual:

Applicable BIS Product Manual for the relevant IS 1785 part.

SIT:

Product-specific mechanical, dimensional and chemical testing.

Required testing:

- Chemical composition
- Tensile strength
- Yield/proof stress where specified
- Elongation
- Relaxation where specified
- Diameter
- Dimensional tolerance
- Surface condition
- Wrapping/bending test where specified
- Other prestressing-wire requirements

Manufacturing / quality requirements:

Controlled wire drawing, heat treatment/stabilization where applicable, surface finishing and mechanical testing are required.

Documents / certification requirements:

- BIS application
- Applicable IS part
- Grade and wire diameter
- Manufacturing-process details
- Heat-treatment records
- Test facilities

- Calibration
- Mechanical-test records
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark together with applicable grade, diameter, manufacturer and traceability information.

Important exclusions:

Prestressing strands and bars are separate products and must not automatically be classified as prestressing wire.

2026 special note:

The AI should distinguish **prestressing wire, strand and bar**, since these are technically different reinforcement products.

Official BIS source:

BIS — Scheme-I Steel and Iron Products

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/scheme-i-mark-scheme/>

CHUNK 43

Product:

Prestressing Steel Strands

Description:

Prestressing strands consist of multiple high-strength steel wires helically laid together and are used to prestress concrete structures. They are widely used in bridges, buildings, railway infrastructure and large concrete structures.

Applicable IS:

IS 6006:2014

Standard title:

Uncoated Stress Relieved Low Relaxation Seven Ply Strands for Prestressed Concrete — Specification

Scope / applicability:

Seven-wire prestressing strands intended for prestressed concrete applications within the specified grades and dimensions.

BIS certification:

Mandatory where covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark.

Applicable QCO:

Steel and Steel Products (Quality Control) Order framework.

Current status — August 2026:

Listed within the applicable compulsory steel-certification framework where covered.

Product Manual:

Applicable BIS Product Manual for IS 6006.

SIT:

Mechanical, dimensional, relaxation and material testing as prescribed.

Required testing:

- Tensile strength
- Yield/proof load
- Elongation
- Relaxation
- Strand diameter
- Nominal cross-sectional area
- Lay characteristics
- Surface condition
- Wire construction
- Breaking load
- Other strand-specific tests

Manufacturing / quality requirements:

Controlled wire production, stranding, stabilization/stress-relieving and finishing processes are required.

Documents / certification requirements:

- BIS application
- Applicable IS
- Strand construction
- Grade
- Manufacturing process
- Heat/batch traceability
- Mechanical-test records
- Relaxation-test records

- Calibration
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark with strand grade, size, manufacturer and traceability details.

Important exclusions:

Prestressing wire and prestressing bars are separate product categories.

2026 special note:

The AI should recognize terms such as ***"prestressing strand," "PC strand" and "7-wire strand"*** but still verify the exact specification.

Official BIS source:

BIS — Products Under Compulsory Certification

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/>

CHUNK 44

Product:

Prestressing Steel Bars

Description:

Prestressing steel bars are high-strength steel bars used to apply prestressing forces to concrete structures. They are used in specialized structural applications where high tensile performance is required.

Applicable IS:

Product-specific BIS standard depends on the type and grade of prestressing bar.

Standard / classification:

The exact IS must be selected according to bar type, surface profile, grade and application.

BIS certification:

Mandatory where covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark where applicable.

Applicable QCO:

Steel and Steel Products (Quality Control) Order framework.

Current status — August 2026:

Exact compulsory scope requires verification.

Product Manual:

To be identified for the applicable prestressing-bar standard.

SIT:

Product-specific chemical, mechanical and dimensional testing.

Required testing:

- Chemical composition
- Tensile strength

- Yield/proof stress
- Elongation
- Relaxation where specified
- Dimensions
- Surface profile
- Straightness
- Hardness where specified
- Other prestressing-specific tests

Manufacturing / quality requirements:

Controlled steelmaking, rolling, heat treatment and surface finishing are required to obtain high-strength prestressing properties.

Documents / certification requirements:

- BIS application
- Applicable IS
- Grade/product range
- Manufacturing process
- Heat-treatment records
- Mechanical tests
- Calibration
- Traceability
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark with grade, size and manufacturer/licence information.

Important exclusions:

Prestressing wire and strand products should not be treated as prestressing bars.

2026 special note:

The AI should ask whether the user means ****bar, wire or strand**** when a query simply says "prestressing steel."

Official BIS source:

BIS — Scheme-I Steel and Iron Products

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/scheme-i-mark-scheme/>

CHUNK 45

Product:

Steel Reinforcement Bars — High Strength Deformed Bars

Description:

High-strength deformed bars are ribbed steel reinforcement bars designed to provide strong mechanical anchorage and high tensile capacity in reinforced concrete. They are extensively used in residential, commercial and infrastructure construction.

Applicable IS:

IS 1786:2008

Standard title:

High Strength Deformed Steel Bars and Wires for Concrete Reinforcement — Specification

Scope / applicability:

High-strength deformed bars and wires intended for reinforcement of concrete structures within the grades and dimensional requirements specified by the standard.

BIS certification:

Mandatory where covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark.

Applicable QCO:

Steel and Steel Products (Quality Control) Order, 2024, as amended.

Current status — August 2026:

Major construction steel product covered under compulsory BIS certification where within QCO scope.

Product Manual:

Applicable BIS Product Manual for IS 1786.

SIT:

Chemical composition, mechanical properties, dimensions, mass and deformation-pattern testing.

Required testing:

- Chemical composition
- Yield strength
- Tensile strength
- Elongation
- Tensile/yield ratio where specified
- Bend test

- Re-bend test
- Mass per metre
- Diameter
- Rib/deformation geometry
- Dimensional tolerances
- Surface condition
- Carbon equivalent where specified

Manufacturing / quality requirements:

Controlled steelmaking, rolling, cooling/heat treatment and rib formation are required to achieve the specified strength and ductility.

Documents / certification requirements:

- BIS application
- Grade range
- Manufacturing-process details
- Heat/batch traceability
- Chemical-analysis records
- Mechanical-test records
- Calibration records
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark together with grade, bar size and manufacturer/licence identification as prescribed.

Important exclusions:

Plain reinforcement bars, prestressing steel and specialized corrosion-resistant reinforcement are separate categories.

2026 special note:

The AI should not answer a query about "TMT bars" only from the commercial name. It should identify the applicable grade and verify IS 1786/QCO requirements.

Official BIS source:

BIS — Products Under Compulsory Certification

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/>

CHUNK 46

Product:

Steel Reinforcement Wire

Description:

Steel reinforcement wire is used in reinforced and prestressed concrete applications and can also be incorporated into welded reinforcement products. Its strength, ductility and dimensional characteristics are important for structural safety.

Applicable IS:

Product-specific BIS standard depends on the reinforcement-wire type.

Standard / classification:

The exact IS must be selected according to whether the wire is plain, deformed, hard-drawn or intended for prestressed concrete.

BIS certification:

Mandatory where covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark where applicable.

Applicable QCO:

Steel and Steel Products (Quality Control) Order framework.

Current status — August 2026:

Exact scope requires product-specific verification.

Product Manual:

To be identified for the applicable reinforcement-wire standard.

SIT:

Mechanical, dimensional and chemical testing according to the applicable standard.

Required testing:

- Chemical composition
- Tensile strength
- Yield/proof stress
- Elongation
- Bend test
- Re-bend where specified
- Diameter
- Mass per metre
- Surface condition
- Deformation geometry where applicable
- Other grade-specific tests

Manufacturing / quality requirements:

Controlled wire drawing, rolling, heat treatment and finishing processes must be maintained.

Documents / certification requirements:

- BIS application
- Applicable IS
- Grade/product range
- Manufacturing-process details
- Testing capability
- Calibration
- Mechanical-test records
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark where applicable with grade, size and manufacturer identification.

Important exclusions:

Prestressing wire and specialized wire products may have dedicated standards.

2026 special note:

The assistant should identify whether the wire is for ordinary reinforcement, welded mesh or prestressed concrete before selecting the standard.

Official BIS source:

BIS — Scheme-I Steel and Iron Products

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/scheme-i-mark-scheme/>

CHUNK 47

Product:

Welded Steel Wire Mesh for Concrete Reinforcement

Description:

Welded steel wire mesh consists of intersecting steel wires welded at their junctions and is used as reinforcement in concrete slabs, walls, pavements and other structural applications. It provides distributed reinforcement and simplifies placement on site.

Applicable IS:

IS 1566:1982

Standard title:

Hard-Drawn Steel Wire Fabric for Concrete Reinforcement — Specification

Scope / applicability:

Steel wire fabric used as reinforcement in concrete construction within the wire sizes, mesh configurations and mechanical requirements specified by the standard.

BIS certification:

Mandatory where covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark.

Applicable QCO:

Steel and Steel Products (Quality Control) Order framework.

Current status — August 2026:

Listed within BIS steel certification material where applicable.

Product Manual:

Applicable BIS Product Manual for IS 1566.

SIT:

Wire mechanical testing, weld/junction inspection, dimensions and mesh geometry.

Required testing:

- Wire tensile strength
- Yield/proof properties where specified
- Elongation
- Weld/junction strength
- Mesh dimensions
- Wire diameter
- Spacing
- Length and width
- Surface condition
- Mass
- Other fabric-specific tests

Manufacturing / quality requirements:

Controlled wire drawing and resistance welding processes are required to maintain consistent wire properties and weld-junction strength.

Documents / certification requirements:

- BIS application
- Applicable IS
- Wire grade
- Mesh configuration
- Manufacturing process
- Welding-process controls
- Testing facilities
- Calibration
- Test records
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark with mesh size, wire diameter, grade and manufacturer/licence identification.

Important exclusions:

Welded wire mesh for fencing and industrial screening is different from reinforcement fabric for concrete.

2026 special note:

The AI should distinguish ****reinforcement mesh**** from ordinary welded wire mesh used for fencing or industrial purposes.

Official BIS source:

BIS — Products Under Compulsory Certification

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/>

CHUNK 48

Product:

Steel Pipes for Structural and General Engineering Applications

Description:

Steel pipes used in structural and general engineering applications provide tubular sections for supports, frames, machinery and fabricated structures. Their strength, dimensions and manufacturing quality determine suitability for the intended load.

Applicable IS:

Product-specific standard depends on pipe construction, grade and intended application.

Standard / classification:

The applicable IS should be selected according to manufacturing method, section, grade and service.

BIS certification:

Mandatory where covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark where applicable.

Applicable QCO:

Steel and Steel Products (Quality Control) Order framework.

Current status — August 2026:

Exact compulsory scope requires product-specific verification.

Product Manual:

To be identified for the applicable structural/general-engineering pipe standard.

SIT:

Chemical, mechanical, dimensional and manufacturing-quality testing.

Required testing:

- Chemical composition
- Tensile strength
- Yield strength
- Elongation
- Bend test
- Flattening test where applicable
- Hydrostatic test where specified
- Dimensions
- Wall thickness
- Straightness
- Weld quality where applicable
- Surface condition
- Other standard-specific tests

Manufacturing / quality requirements:

Controlled forming, welding, heat treatment where applicable, sizing and finishing processes are required.

Documents / certification requirements:

- BIS application
- Applicable IS

- Manufacturing-process details
- Grade/product range
- Welding/process controls
- Testing capability
- Calibration
- Test records
- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark with size, grade and manufacturer/licence information.

Important exclusions:

Pipes intended specifically for boilers, oil/gas pipelines, water wells or potable water may be covered by separate standards.

2026 special note:

"MS pipe" is not sufficient information for a definitive BIS answer. The AI should determine the intended application and pipe specification first.

Official BIS source:

BIS — Products Under Compulsory Certification

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/>

CHUNK 49

Product:

Steel Drum and Barrel Material / Steel Containers

Description:

Steel containers such as drums and barrels are used for storing and transporting liquids, chemicals, oils, food-related materials and other goods. Material strength, welding quality, dimensional accuracy and corrosion protection are important for safe containment.

Applicable IS:

Product-specific BIS standard depends on container type, capacity, material and intended contents.

Standard / classification:

The exact IS must be identified according to whether the container is a closed-head drum, open-head drum, barrel or another steel container.

BIS certification:

Mandatory where the exact product is covered by the applicable QCO.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark where applicable.

Applicable QCO:

Steel and Steel Products (Quality Control) Order framework where applicable.

Current status — August 2026:

Exact standard and compulsory scope require verification.

Product Manual:

To be identified after confirming the exact steel-container standard.

SIT:

Product-specific dimensional, mechanical, leakage and performance testing.

Required testing:

- Sheet thickness
- Dimensions
- Capacity
- Weld quality
- Leakage/tightness
- Drop/handling resistance where specified
- Closure performance
- Corrosion protection/coating
- Surface condition
- Material properties
- Other container-specific tests

Manufacturing / quality requirements:

Controlled sheet forming, seam welding, closure fitting, coating and leak-testing processes are required.

Documents / certification requirements:

- BIS application
- Applicable IS
- Container type/capacity
- Material details
- Manufacturing process
- Welding controls
- Test facilities
- Calibration
- Test records

- Product Manual/SIT
- Factory assessment

Marking requirements:

BIS Standard Mark / ISI Mark where applicable with capacity, product identification and manufacturer/licence information.

Important exclusions:

Containers for dangerous goods, food contact or specialized industrial chemicals may have additional regulatory requirements beyond BIS.

2026 special note:

The AI should identify the container's contents and intended use because packaging requirements can differ significantly between ordinary industrial storage and regulated goods.

Official BIS source:

BIS — Products Under Compulsory Certification

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/>

CHUNK 50

Product:

Steel Slag / Steelmaking By-products for Construction Use

Description:

Steel slag is a by-product of steelmaking that can be processed and used in selected construction applications. Because its properties depend heavily on production and

processing, chemical stability, expansion and other performance characteristics require careful control.

Applicable IS:

Product-specific BIS standard depends on the intended construction application.

Standard / classification:

The applicable IS must be selected according to whether the material is used as aggregate, road material, cement-related material or another application.

BIS certification:

Current compulsory BIS status requires product-specific verification.

Certification scheme:

Scheme-I — BIS Standard Mark / ISI Mark where applicable if covered by the applicable QCO.

Applicable QCO:

Steel and Steel Products (Quality Control) Order framework where applicable to the exact product.

Current status — August 2026:

Scope verification required.

Product Manual:

To be identified for the exact steel-slag product standard.

SIT:

Depends on the intended construction application.

Required testing:

- Chemical composition
- Free lime
- Free magnesia where specified
- Volume stability/expansion
- Particle size distribution
- Density
- Water absorption
- Soundness where applicable
- Strength-related properties
- Leaching/environmental parameters where specified
- Other application-specific tests

Manufacturing / quality requirements:

Slag processing, ageing/stabilization, crushing, screening and removal of metallic contaminants must be controlled according to the intended use.

Documents / certification requirements:

- BIS application where applicable
- Applicable IS
- Production/process details
- Processing records
- Laboratory facilities
- Test records
- Calibration
- Product Manual/SIT
- Factory assessment where applicable

Marking requirements:

BIS Standard Mark only where applicable and certified.

Important exclusions:

Steel slag intended for specialized industrial applications may not have the same requirements as slag used as construction aggregate.

2026 special note:

The AI should treat steel slag as an application-specific material rather than assuming one universal BIS standard. The intended end use must be established first.

Official BIS source:

BIS — Products Under Compulsory Certification

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/>